

Database Core Concepts

In SQL terminology:

Relational Theory Term	SQL Term
Relation	Table
Tuple	Row
Attribute	Column
Cell	Individual Value

A table is a logical database object consisting of rows and columns that stores information about a particular entity or concept.

Logical Table

This is how users think about the data. You simply see rows and columns.

You don't care:

- Where it's stored or how its stored .

Physical Table

Internally, the DBMS stores the same data using:

- Pages
- Blocks
- Records
- Buffer pool
- B+ Trees
- Heap files .

Execution Perspective

When you execute:

```
SELECT * FROM Students;
```

You think:

"Read the Students table."

The DBMS thinks:

"Locate the relevant pages, choose an index scan or sequential scan, fetch records into memory, apply predicates, and return the result."

Definitions: Rows, Columns, and Cells

What is a Row?

A row represents one complete record. A tuple is a single element of a relation consisting of one value for every attribute. ordered collection of attribute values is a tuple. when we say tuples are unordered it means , there is no imposition of order on its rows .

What is a Column?

A column stores one type of information for all records . An attribute is a property or characteristic of an entity represented within a relation.

Example: Student entity

Properties:

- StudentID
- Name
- CGPA
- Email .

every attribute has name , domain , data type , meaning .

Cell

cell is the smallest logical unit of data in a relation and contains exactly one value for a specific tuple and attribute. It must contain one atomic value so that the relation satisfies the requirements of the relational model and First Normal Form.

Theory vs. Implementation

tuples and attributes are relational theory terms , rows and columns are sql implementation and represents the actual table stored in dbms .

Relations are defined as sets of tuples, and sets are inherently unordered. Therefore, the sequence in which tuples appear has no logical meaning. SQL guarantees a specific row order only when an ORDER BY clause is used.